

Regime-Aware Machine Learning for Technical Pattern Trading

Event-Based Classification and Profitability Evaluation on SPY

Zeineb Turki

Supervisor: [Supervisor Name]

[University / Course Name]

Summer Semester 2026

Problem Motivation

Why financial ML needs event-based approaches

- **Financial markets are extremely noisy** — most bars contain no actionable signal
- **Naive price prediction** models learn noise, price level, or time trends
- **Event-based learning** focuses on *technically meaningful moments*

Approach	Prediction Target	Key Problem
Naive regression	Next bar's close price	Learns noise & trends
Bar-level classification	Up/down per bar	No trading context
Event-based (this work)	Trade outcome at signal bars	Focused, actionable

Core question: *When price is at a technically meaningful structure, can we predict the outcome of a trade-sized move?*

Background

Foundations of the approach

- **Technical analysis patterns:** support/resistance, channels, triangles, multiple tops/bottoms
- **Triple-barrier labeling** (Lopez de Prado, 2018): labels based on trade outcomes, not arbitrary direction
- **Random Forest / Bagging:** ensemble tree models — robust, interpretable, handle mixed features
- **Walk-forward validation:** simulates realistic deployment with expanding training windows
- **Profitability evaluation:** backtesting with TP/SL/time exits — accuracy alone is insufficient

Key insight from financial ML literature: the best classifier is not necessarily the best trading model. Profitability depends on the *magnitude* of correct predictions, not just their frequency.

System Architecture

End-to-end pipeline from raw data to profitability evaluation



Stage	Module	Output
Data	load_data.py	4,023 SPY daily bars (2010-2025)
Detection	scanner.py + 4 detectors	132 event signals
Features	indicators.py + build_features.py	48 features per event
Labeling	label_events.py	long / short / no_trade labels
Training	train.py + optimize.py	RF, Bagging, Baseline models
Evaluation	simulator.py + train.py	Classification + profitability metrics

Pattern Detection

Four complementary detectors identify trading opportunities

Detector	Method	Signals	Key Parameters
Support / Resistance	Rolling max/min + stability filter	42 events	window=50, $0.3 \times \text{ATR}$, cooldown=10
Triangles	Swing pivots + linear regression	17 events	window=25, $ r \geq 0.85$, convergence $\geq 5\%$
Channels	Chunk extremes + polyfit	12 events	lookback=40 $\pm$ 15, $\geq 2+3$ touches
Multiple Tops/Bottoms	Rolling extremes + trend confirm	63 events	window=20, 5-bar slope confirm
TOTAL		132 events	3.3% event density

- All detectors use **cooldown filters** (10 bars) to prevent signal clustering
- Signal localized at the **event bar** — the bar where the pattern condition is met
- Per supervisor feedback: triangles and channels excluded from training labels

Feature Engineering

48 features with strict leakage prevention

Feature Group	Count	Examples
Volatility & Returns	9	atr_ratio, rvol_20, ret_1/5/10/20, mom_5/10/20
Moving Average Distances	8	sma_10_dist ... sma_200_dist, MA spreads
Momentum Indicators	4	rsi_14, macd_norm, macd_signal_norm, macd_hist_norm
Bollinger Bands	2	bb_width, bb_pctb
Binary Technical Filters	8	bb_touch_upper/lower, SMA cross, RSI extremes
Volume	2	volume_ratio, volume_std
Pattern Geometry	11	slopes, touches, containment, width, R ²
Event Type Dummies	4–6	One-hot encoded pattern type
TOTAL	48–50	

Removed (trend-leaking): raw ATR, absolute SMA values, cumulative OBV, raw MACD. All features use relative, normalized, or bounded values only.

Triple-Barrier Labeling & Optimization

TP/SL/max_holding treated as tunable hyperparameters

- **Triple-barrier method:** for each event, walk forward and check barriers

Barrier	Definition	Label if hit first
Upper (TP)	$\text{entry_price} + \text{pt_mult} \times \text{ATR}$	long
Lower (SL)	$\text{entry_price} - \text{sl_mult} \times \text{ATR}$	short
Time	max_holding bars elapsed	no_trade

Innovation: pt_mult, sl_mult, max_holding are *not fixed* — they are optimized via grid search (100 configurations).

Search Space	Range	Best for F1	Best for Profit
pt_mult	1.0 – 3.0	2.0	2.5
sl_mult	1.0 – 3.0	1.5	3.0
max_holding	5 – 20 bars	10	20
Score		F1 = 0.569	Return = 25.9%

Machine Learning & Validation

Three models, three validation strategies

- **Models:** Random Forest (200 trees), Bagging (200 trees), Stratified Baseline
- **Class weighting:** balanced (adjusts for label imbalance)

Validation Method	Description	Purpose
Chronological split	60% train / 20% val / 20% test	Primary evaluation
Walk-forward CV	5 expanding folds, temporal order	Realistic deployment simulation
5-fold event CV	Rotate test fold, 80/20 train/val	Reduce sampling noise (diagnostic)
Tree diagnostics	Per-tree accuracy, ensemble gain	Detect ensemble dependency

- Walk-forward CV respects temporal order — no future data in training
- Individual tree analysis shows whether ensemble relies on few strong trees or distributes strength

Trading Simulation & Profitability

Why accuracy alone is insufficient

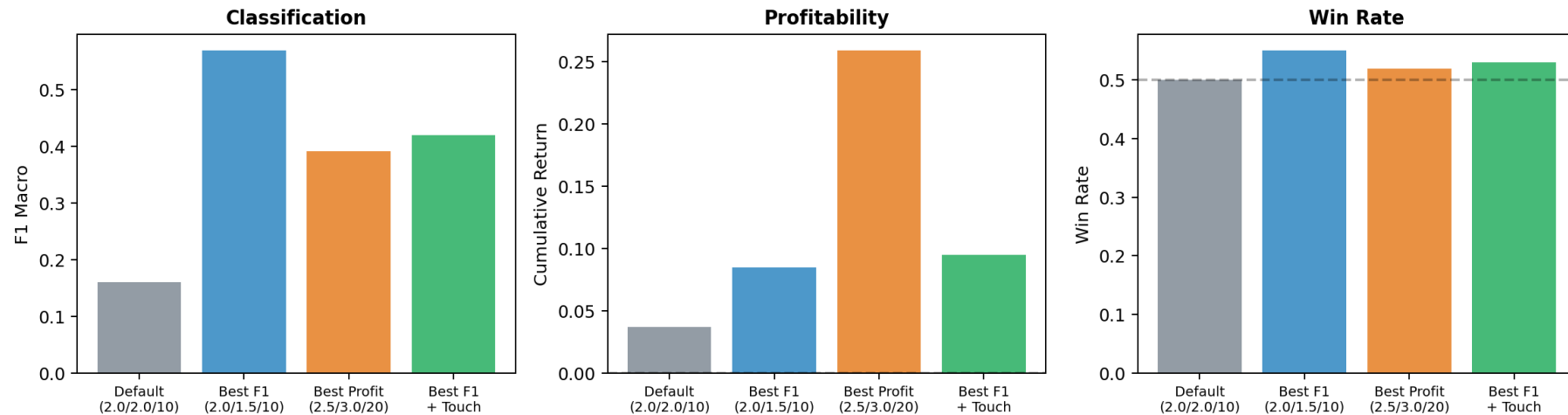
- **Simulated trading:** enter at signal-bar close, exit at TP/SL/time barrier
- A model with **lower accuracy** can be **more profitable** if correct predictions capture larger moves

Metric	Definition	Interpretation
Cumulative return	$\Sigma(\text{net returns})$	Total P&L
Win rate	% trades with positive return	Hit rate
Profit factor	Gross profit / gross loss	Reward-to-risk ratio
Sharpe ratio	Mean return / std(returns)	Risk-adjusted performance
Max drawdown	Largest peak-to-trough decline	Worst-case scenario

Assumptions: no transaction costs, no slippage, equal position sizing, no compounding. Entry at signal-bar Close (consistent with labeling pipeline).

Experimental Results

104 detector events + 38 touch events = 142 total



Configuration	pt / sl / mh	F1 Macro	Cum. Return	Win Rate	Trades
Default (baseline)	2.0 / 2.0 / 10	0.160	3.7%	50%	18
Best for F1	2.0 / 1.5 / 10	0.569	8.5%	55%	18
Best for Profit	2.5 / 3.0 / 20	0.392	25.9%	52%	15
Best F1 + Touch	2.0 / 1.5 / 10	0.420	9.5%	53%	24

Key finding: optimal parameters for classification (F1) differ from those for profitability (return). This confirms that trading evaluation requires profitability metrics beyond accuracy.

Generalization & Variance Analysis

How stable are results across time folds?

- Walk-forward CV (5 folds): **mean $\pm$ std** reported for all metrics
- High variance across folds = instability — results depend on which period is tested

Metric	WF Mean	WF Std	Interpretation
F1 Macro	≈ 0.27	≈ 0.03	Moderate stability
Cum. Return	≈ 0.03	≈ 0.03	High relative variance
Win Rate	$\approx 50\%$	$\approx 7\%$	Thin edge, regime-dependent
Sharpe	≈ 0.11	≈ 0.14	Unstable risk-adjusted return

Key insight: point estimates like 'F1=0.569' represent one realisation from a high-variance distribution. The true out-of-sample performance could be substantially different.

F-Beta Analysis

Precision vs recall for trading decisions

- **F0.5 (precision-heavy)**: fewer but cleaner trades — avoid false positives
- **F1.0 (balanced)**: equal weight to false positives and false negatives
- **F2.0 (recall-heavy)**: capture more opportunities — tolerate false positives

Error Type	Trading Meaning	Cost
False Positive	Model says trade, but it loses	Direct financial loss
False Negative	Model says skip, but it would have won	Missed opportunity

Finding: Precision remains low across all folds, meaning many signals are false alarms. The optimal beta depends on risk tolerance: capital-preservation strategies should favour F0.5, opportunity-seeking strategies should favour F2.0.

Limitations

Honest assessment of constraints

- **Small dataset:** ~140 events provides limited statistical power; high result variance
- **Single asset:** only SPY tested — generalization to other markets is unknown
- **No transaction costs:** spread, slippage, and commissions are not modeled
- **Simplified entry:** signal-bar Close, not next-bar Open (slight optimism)
- **Optimization overfitting risk:** 100 configurations on ~100 events
- **Touch-event noise:** additional events may not carry the same signal quality as strict detectors
- **No regime modeling:** HMM regime detection planned but not yet integrated

All results should be interpreted as *preliminary research findings*, not production-ready trading signals.

Conclusion & Future Work

What was achieved:

- End-to-end pipeline: data → detection → features → labeling → ML → validation → backtest
- 48 leakage-free features from 4 pattern detectors on 4,023 SPY bars
- Triple-barrier parameters converted from fixed constants to tunable hyperparameters
- Profitability evaluation integrated into training and cross-validation
- 3.6× F1 improvement over baseline; 25.9% cumulative return (best profit config)

Main scientific lesson:

- Classification accuracy and trading profitability optimize at *different* parameter settings

Future work:

- Multi-asset testing, transaction cost modeling, purged/embargo CV
- Regime-aware dynamic TP/SL, HMM integration, more advanced models